

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR SYNTHESIS AND COMPOSITION CONTROL

1. ABSTRACT

This report synthesizes three primary patents relevant to low acrylonitrile nitrile butadiene rubber (NBR) and adjacent process technologies. The target material is NBR with reduced acrylonitrile (ACN) content relative to conventional grades, typically to improve low-temperature flexibility while retaining oil resistance. The patent set spans two direct NBR preparation routes and one enabling metathesis technology that supports downstream molecular-weight adjustment of NBR prior to hydrogenation or property tuning. EP2316860B1 discloses emulsion polymerization of at least one α,β -unsaturated nitrile with at least one conjugated diene using a redox initiator system containing a peroxy oxidant, a reducing agent, and Fe(II) chloride; the specification emphasizes conventional emulsion NBR manufacture and the use of metal-salt/redox initiation. US9023914B2 teaches free-radical solution polymerization of NBR in organic solvents with specific modifier substances, enabling control of molecular architecture, molecular weight, and polydispersity; its examples include acrylonitrile/butadiene feed ratios such as 37/63 phm and a lower-ACN terpolymer at 30/50 phm ACN/butadiene. US8507398B2 is not a direct NBR synthesis patent, but it is highly relevant because it provides ruthenium- or osmium-carbene metathesis catalysts for metathesis of nitrile rubber, supporting post-polymerization chain modification or molecular-weight reduction strategies that can be important when tailoring low-ACN NBR for hydrogenation or viscosity control. Across the landscape, the dominant process themes are emulsion versus solution polymerization, redox or radical initiation, use of chain-transfer/modifier chemistry for molecular-weight control, and the possibility of subsequent hydrogenation or metathesis-based adjustment. The evidence supports low-ACN NBR synthesis by controlled copolymerization of butadiene with acrylonitrile, with process selection strongly influencing polymer architecture, viscosity, and downstream processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2316860B1
- Patent Title: Nitrile rubbers

- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses emulsion polymerization of at least one α,β -unsaturated nitrile with at least one conjugated diene, optionally with further copolymerizable monomers, to produce nitrile rubber. The key feature is a redox initiation system comprising a peroxo oxidant, a reducing agent, and Fe(II) chloride; the patent also discusses conventional NBR/HNBR preparation and optional hydrogenation of the resulting nitrile rubber. - Nitrile content: 10-60 wt% unsaturated nitrile — cited discussion of prior art nitrile rubbers - Mooney viscosity: 15-150 — cited discussion of prior art nitrile rubbers - Mooney viscosity: 15-65 — cited discussion of EP-A-0 692 496 prior art - Hydrogenation degree: 50-100% — detailed description of HNBR - Polymerization temperature: 20-30 °C — cited discussion of US-A-2,451,180 prior art - Initiator system: peroxo compound + reducing agent + Fe(II) chloride — claims - Oxidizing agent: hydrogen peroxide — claim 2 - Oxidizing agent: peroxodisulphates — claim 2 - Oxidizing agent: peroxodiphosphates — claim 2 - Oxidizing agent: hydroperoxides — claim 2 - Oxidizing agent: peracids — claim 2 - Oxidizing agent: peracid esters — claim 2 - Oxidizing agent: peracid anhydrides — claim 2 - Oxidizing agent: peroxides having two organic radicals — claim 2 - Oxidizing agent: sodium, potassium or ammonium salts of peroxodisulphuric acid or peroxodiphosphoric acid — claim 3 - Oxidizing agent: t-butyl hydroperoxide — claim 3 - Oxidizing agent: cumene hydroperoxide — claim 3 - Oxidizing agent: pinane hydroperoxide — claim 3 - Oxidizing agent: p-menthane hydroperoxide — claim 3 - Oxidizing agent: dibenzoyl peroxide — claim 3 - Oxidizing agent: 2,4-dichlorobenzoyl peroxide — claim 3 - Oxidizing agent: di-t-butyl peroxide — claim 3 - Oxidizing agent: dicumyl peroxide — claim 3 - Oxidizing agent: t-butyl perbenzoate — claim 3 - Oxidizing agent: t-butyl peracetate — claim 3 - Reducing agent: sodium formaldehyde-sulphoxylate — claim 4 - Reducing agent: sodium benzaldehyde-sulphoxylate — claim 4 - Reducing agent: reducing sugars — claim 4 - Reducing agent: ascorbic acid — claim 4 - Reducing agent: sulphenates — claim 4 - Reducing agent: sulphinates — claim 4 - Reducing agent: sulfoxylates — claim 4 - Reducing agent: dithionite — claim 4 - Reducing agent: sulphite — claim 4 - Reducing agent: metabisulphite — claim 4 - Reducing agent: disulphite — claim 4 - Reducing agent: sugars — claim 4 - Reducing agent: urea — claim 4 - Reducing agent: thiourea — claim 4 - Reducing agent: xanthogenates — claim 4 - Reducing agent: thioxanthogenates — claim 4 - Reducing agent: hydrazinium salts — claim 4 - Reducing agent: amines and amine derivatives — claim 4 - Preferred reducing agent examples: aniline, dimethylaniline, monoethanolamine, dieth... — claim 4 - Chain transfer agent: C12-C16 alkylthiol — cited prior art discussion - Chain transfer agent loading: at least 0.03 mol per 100 mol monomer units — cited prior art discussion

Relevant Experimental Evidence - Example 1: Not structurally extracted by parser; the patent text mainly establishes the redox emulsion polymerization concept rather than a worked example

set. - Example 2: Not detected by parser. - Example 3: Not detected by parser. - Example 4: Not detected by parser. - Example 5: Not detected by parser.

Technical Relevance This patent is directly relevant because it discloses the standard emulsion route for NBR manufacture and a specific redox initiation system that can be applied to acrylonitrile/butadiene copolymerization. Although it does not focus specifically on low-ACN grades, it provides a practical synthesis framework for NBR production and composition control.

Patent 2

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical solution polymerization of nitrile rubbers in organic solvents using specific modifier substances, followed optionally by hydrogenation. The process is designed to incorporate modifier fragments into the main chain and/or chain ends, enabling control of molecular weight and polydispersity while preparing NBR and related terpolymers. - Acrylonitrile: 37 phm — Example 1/Table 1 - 1,3-Butadiene: 63 phm — Example 1/Table 1 - Dimethylacetamide: 51 phm — Example 1/Table 1 - DoPAT: 0.22 phm — Example 1/Table 1 - Vazo® 88: 0.310 phm — Example 1/Table 1 - Pyrocatechol/hydroquinone: 5.7 phm — Example 1/Table 1 - Polymerization temperature: 100 °C — Example 1/Table 1 - Polymerization time: 9 h — Example 1/Table 1 - Final conversion: 50% — Example 1/Table 1 - Acrylonitrile: 30 phm — Example 6/Table 1 - 1,3-Butadiene: 50 phm — Example 6/Table 1 - Hydroxyethyl acrylate: 20 phm — Example 6/Table 1 - DoPAT: 0.197 phm — Example 6/Table 1 - Vazo® 88: 0.275 phm — Example 6/Table 1 - Dimethylacetamide: 45 phm — Example 6/Table 1 - Polymerization temperature: 100 °C — Example 6/Table 1 - Polymerization time: 6 h — Example 6/Table 1 - Final conversion: 39% — Example 6/Table 1 - DoPAT (NBR #7): 2.22 phm — Example 7/Table 1 - Acrylonitrile: 37 phm — Example 7/Table 1 - 1,3-Butadiene: 63 phm — Example 7/Table 1 - Vazo® 88: 0.310 phm — Example 7/Table 1 - Polymerization temperature: 100 °C — Example 7/Table 1 - Polymerization time: 9 h — Example 7/Table 1 - Final conversion: 48% — Example 7/Table 1 - Styrene: 50 phm — Example 8/Table 1 - NBR #7 macromer: 50 phm — Example 8/Table 1 - DoPAT-NBR#7: 50 phm — Example 8/Table 1 - Vazo® 88: 0.056 phm — Example 8/Table 1 - Dimethylacetamide: 63 phm — Example 8/Table 1 - Polymerization temperature: 100 °C — Example 8/Table 1 - Polymerization time: 13 h — Example 8/Table 1 - Final conversion: 9.2% — Example 8/Table 1

Relevant Experimental Evidence - Example 1: Demonstrated solution polymerization of NBR from acrylonitrile and 1,3-butadiene at 37/63 phm in DMAc at 100 °C, giving 50% conversion after

9 h. - Example 6: Demonstrated a lower-ACN terpolymer containing 30 phm acrylonitrile and 50 phm butadiene with hydroxyethyl acrylate as termonomer, reaching 39% conversion in 6 h. - Example 7: Demonstrated deliberate preparation of a low-molecular-mass NBR by increasing modifier concentration to 2.22 phm, reaching 48% conversion after 9 h. - Example 8: Demonstrated use of the NBR #7 macromer in a block-styrene terpolymer synthesis at 100 °C, showing the utility of solution-made NBR as a reactive intermediate. - Example 2: The example series indicates that modifier concentration and solvent choice were varied to tune polymer properties, though the parser did not extract a separate structured record.

Technical Relevance This patent is highly relevant because it directly discloses NBR synthesis in organic solvent and includes a lower-acrylonitrile formulation (30 phm ACN) that aligns with the low-ACN target. It also shows how modifier concentration can be used to control molecular weight and architecture, which is important for processing and low-temperature performance.

Patent 3

- Patent Number: US8507398B2
- Patent Title: Catalysts for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses ruthenium- or osmium-carbene metathesis catalysts for metathesis reactions, including metathesis of nitrile rubber. The technology is not a primary NBR polymerization route, but it enables post-polymerization chain modification, depolymerization, or molecular-weight adjustment of nitrile rubber materials. - Time: 5 min — Example 1/A2 - Conversion: 53.5% — Example 1/A2 - Conversion: 68.2% — Example 1/A2 - Conversion: 96.3% — Example 1/A2 - Conversion: 74.7% — Example 1/A2 - Conversion: 92.3% — Example 1/A2 - Conversion: 93.9% — Example 1/A2 - Conversion: 92.9% — Example 1/A2 - Conversion: 83.4% — Example 1/A2 - Conversion: 89.8% — Example 1/A2 - Time: 10 min — Example 1/A2 - Conversion: 68.0% — Example 1/A2 - Conversion: 79.0% — Example 1/A2 - Conversion: 99.0% — Example 1/A2 - Conversion: 86.9% — Example 1/A2 - Conversion: 96.4% — Example 1/A2 - Conversion: 97.9% — Example 1/A2 - Conversion: 96.8% — Example 1/A2 - Conversion: 90.4% — Example 1/A2 - Conversion: 94.2% — Example 1/A2 - Time: 15 min — Example 1/A2 - Conversion: 78.1% — Example 1/A2 - Conversion: 84.7% — Example 1/A2 - Conversion: 99.7% — Example 1/A2 - Conversion: 94.3% — Example 1/A2 - Conversion: 97.6% — Example 1/A2 - Conversion: 99.2% — Example 1/A2 - Conversion: 98.1% — Example 1/A2 - Conversion: 93.7% — Example 1/A2 - Conversion: 96.3% — Example 1/A2 - Time: 20 min — Example 1/A2 - Conversion: 86.8% — Example 1/A2 - Conversion: 88.5% — Example 1/A2 - Conversion: 96.4% — Example 1/A2 - Conversion: 98.2% — Example 1/A2 - Conversion: 99.5% — Example 1/A2 - Conversion: 98.7% — Example 1/A2 -

Conversion: 95.1% — Example 1/A2 - Conversion: 97.2% — Example 1/A2 - Time: 25 min — Example 1/A2 - Conversion: 91.3% — Example 1/A2 - Conversion: 90.8% — Example 1/A2 - Conversion: 97.8% — Example 1/A2 - Conversion: 98.3% — Example 1/A2 - Conversion: 99.7% — Example 1/A2 - Conversion: 99.0% — Example 1/A2 - Conversion: 96.0% — Example 1/A2 - Conversion: 97.7% — Example 1/A2 - Time: 30 min — Example 1/A2 - Conversion: 94.2% — Example 1/A2 - Conversion: 92.8% — Example 1/A2 - Conversion: 98.6% — Example 1/A2 - Conversion: 98.4% — Example 1/A2 - Conversion: 99.1% — Example 1/A2 - Conversion: 96.5% — Example 1/A2 - Conversion: 98.0% — Example 1/A2 - Time: 45 min — Example 1/A2 - Conversion: 97.8% — Example 1/A2 - Conversion: 95.8% — Example 1/A2 - Conversion: 99.4% — Example 1/A2 - Conversion: 98.4% — Example 1/A2 - Conversion: 99.3% — Example 1/A2 - Conversion: 97.4% — Example 1/A2 - Conversion: 98.5% — Example 1/A2 - Time: 60 min — Example 1/A2 - Conversion: 99.1% — Example 1/A2 - Conversion: 97.4% — Example 1/A2 - Conversion: 99.6% — Example 1/A2 - Conversion: 98.5% — Example 1/A2 - Conversion: 99.5% — Example 1/A2 - Conversion: 97.6% — Example 1/A2 - Conversion: 98.8% — Example 1/A2 - DEDAM amount: 0.151 ml (0.625 mmol) — Example A3 - Chlorobenzene: 0.3 ml — Example A3 - CDCl₃: 0.2 ml — Example A3 - CaCl₂: 1 mg — Example A3 - Reaction start temperature: room temperature — Example A3 - Catalyst amount: 15.4 mg — Example A3 - Catalyst amount: 11.05×10^{-6} mol — Example A3

Relevant Experimental Evidence - Example 1: Showed rapid metathesis conversion over 5-60 min for multiple catalyst systems, with several catalysts reaching near-complete conversion. - Example A2: Demonstrated ring-closing metathesis of diallylmalononitrile under inert gas, comparing catalyst performance across time points. - Example A3: Demonstrated ring-closing metathesis of diethyl diallylmalonate under aerobic conditions and showed that CaCl₂ can positively influence the reaction. - Example 1: The catalyst set included Grubbs II, Grubbs III, and catalysts K2-K5, establishing broad metathesis utility. - Example A3: The reaction was initiated at room temperature, supporting operationally simple metathesis conditions.

Technical Relevance This patent is relevant because metathesis can be used to modify nitrile rubber chain length and architecture, which is useful when tuning low-ACN NBR viscosity or preparing hydrogenation feedstocks. While it does not synthesize NBR directly, it supports downstream processing strategies that complement low-ACN NBR manufacture.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2316860B1 and US9023914B2 both address direct NBR formation, but EP2316860B1 is an emulsion process using a peroxy/reducing-agent/Fe(II) chloride redox system, whereas US9023914B2 uses free-radical solution polymerization in organic solvent with specific modifiers.
- US9023914B2 explicitly discloses a lower-ACN formulation at 30 phm acrylonitrile and 50 phm butadiene, while EP2316860B1 emphasizes general NBR composition ranges rather than a

specific low-ACN recipe.

- EP2316860B1 relies on emulsion polymerization and conventional chain-transfer/initiator concepts from prior art, whereas US9023914B2 emphasizes organic-solvent synthesis and modifier incorporation to control molecular weight and polydispersity.
- US8507398B2 is not a polymerization patent, but it is process-enabling for NBR by providing metathesis catalysts that can be used for chain modification or molecular-weight reduction before or after hydrogenation.
- Across the set, temperature control is explicit in US9023914B2 at 100 °C and in EP2316860B1's cited prior art at 20-30 °C; US8507398B2 demonstrates metathesis at room temperature in one example and over 5-60 min time windows.
- Only US9023914B2 directly demonstrates a composition shift toward lower acrylonitrile content; the other patents mainly contribute process control or downstream modification tools.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR is most directly achieved by controlled copolymerization of butadiene with acrylonitrile, with process choice strongly affecting molecular weight and processability. Among the primary patents, US9023914B2 is the most directly useful for low-ACN synthesis because it discloses solution polymerization with an explicit lower-ACN example (30 phm ACN, 50 phm butadiene) and modifier-based control of polymer architecture. EP2316860B1 supports conventional emulsion NBR manufacture using a redox/Fe(II) chloride initiation system, which is industrially relevant for NBR production but less specific to low-ACN composition tuning. US8507398B2 adds a complementary metathesis toolkit for post-polymerization molecular-weight adjustment and chain modification. Based on the evidence, the most defensible synthesis strategy for low-ACN NBR is controlled butadiene/acrylonitrile copolymerization, with acrylonitrile held at the lower end of the NBR range and molecular weight managed by modifier or chain-transfer chemistry; solution polymerization offers the clearest disclosed route for architecture control, while emulsion polymerization remains the standard industrial route.

5. REFERENCES

- EP2316860B1 | Nitrile rubbers | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2316860B1/en>
- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- US8507398B2 | Catalysts for metathesis reactions | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8507398B2/en>